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REDUCING CORRUPTION TROUGH BLOCKCHAIN-BASED TRANSPARENCY SYSTEMS

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INTRODUCTION

- Corruption remains a major issue in Indonesian governance.
- Centralized information systems are vulnerable to data manipulation.
- Weak transparency reduces accountability and public trust.
- Existing systems rely heavily on administrative honesty.
- A transparent and tamper-proof system is required.

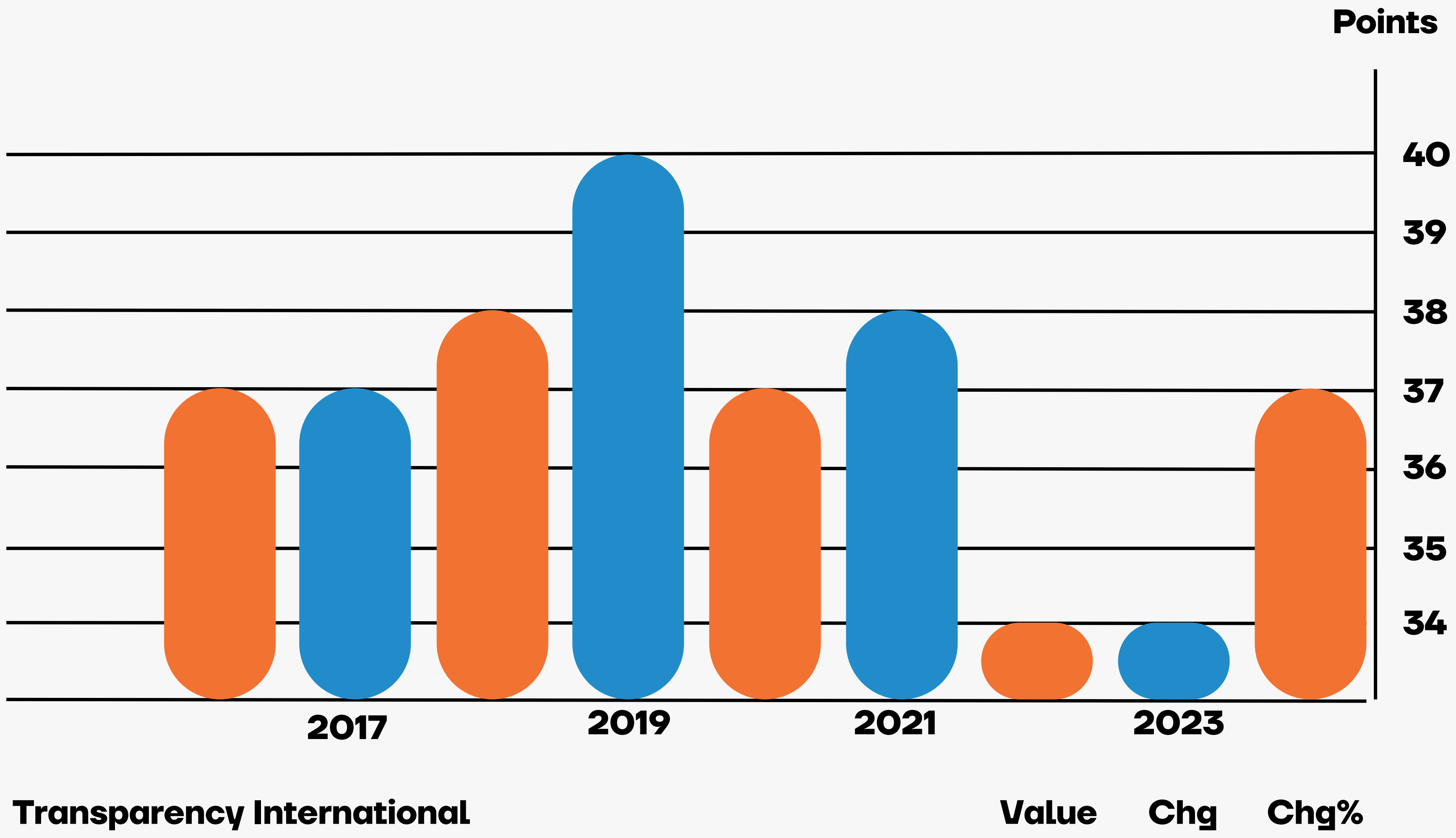

Fig. 1. Corruption Index in Indonesia

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RESEARCH OBJECTIVES

- To analyze the limitations of centralized data systems.
- To design a blockchain-based transparency system.
- To compare centralized and blockchain-based architectures.
- To evaluate data integrity, transparency, and auditability

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LITERATURE REVIEW OVERVIEW

1. Corruption involves abuse of authority for personal or institutional gain.
2. Transparency is essential for accountability and trust.
3. Blockchain provides immutability, transparency, and security.
4. Previous studies show blockchain’s potential in public governance.

BLOCKCHAIN CONCEPT & ARCHITECTURE

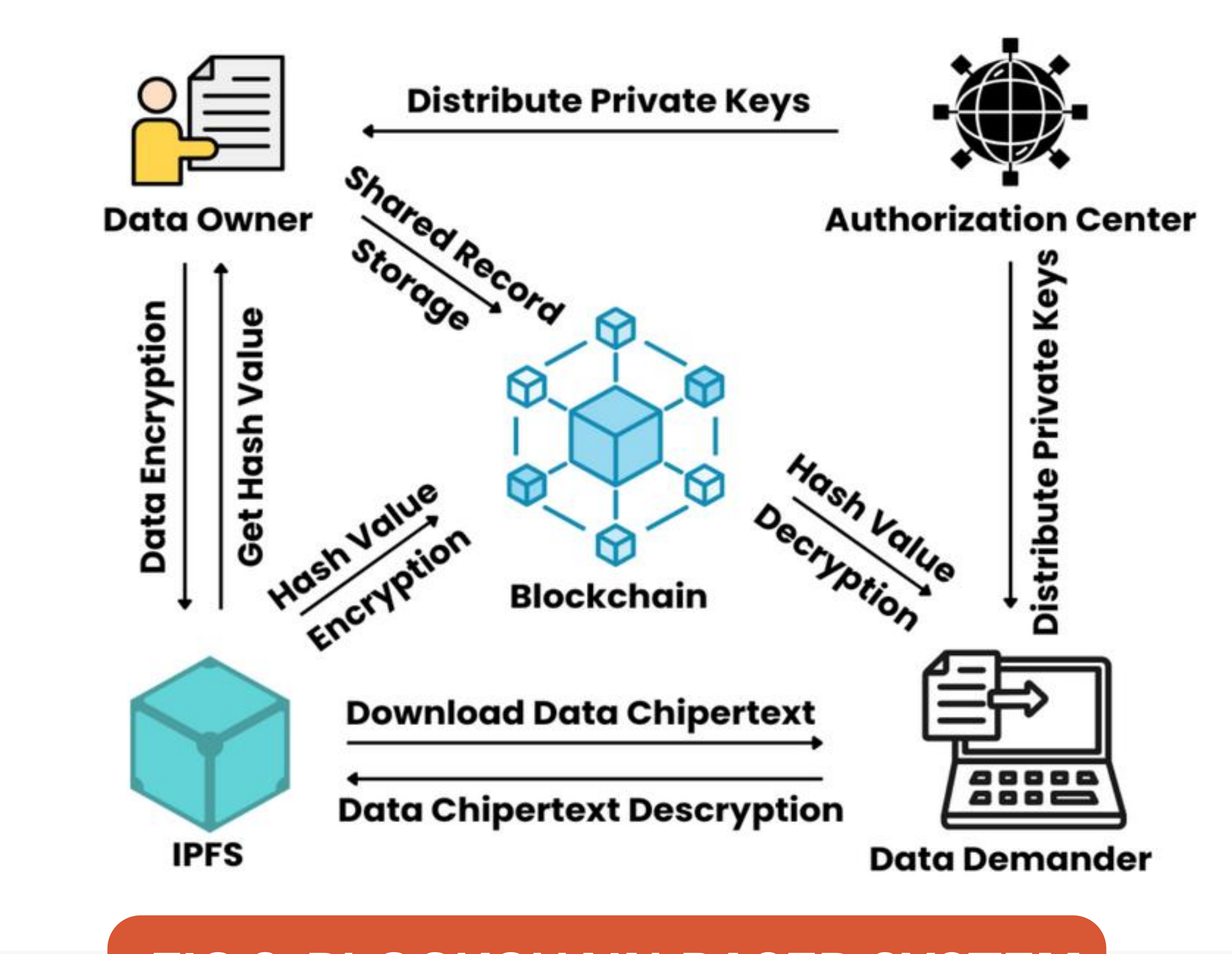

FIG 2. BLOCKCHAIN-BASED SYSTEM

1. Blockchain is a decentralized ledger system.
2. Transactions are cryptographically secured and immutable.
3. No single authority can alter recorded data.
4. Ensures public verifiability and traceability.

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RESEARCH METHODOLOGY

1. Experimental research design.
2. Two prototype systems developed:
   - System A: Centralized system
   - System B: Blockchain-based system
1. Both systems tested under identical conditions.
2. Performance compared using transparency and integrity metrics.

SYSTEM A: CENTRALIZED SYSTEM

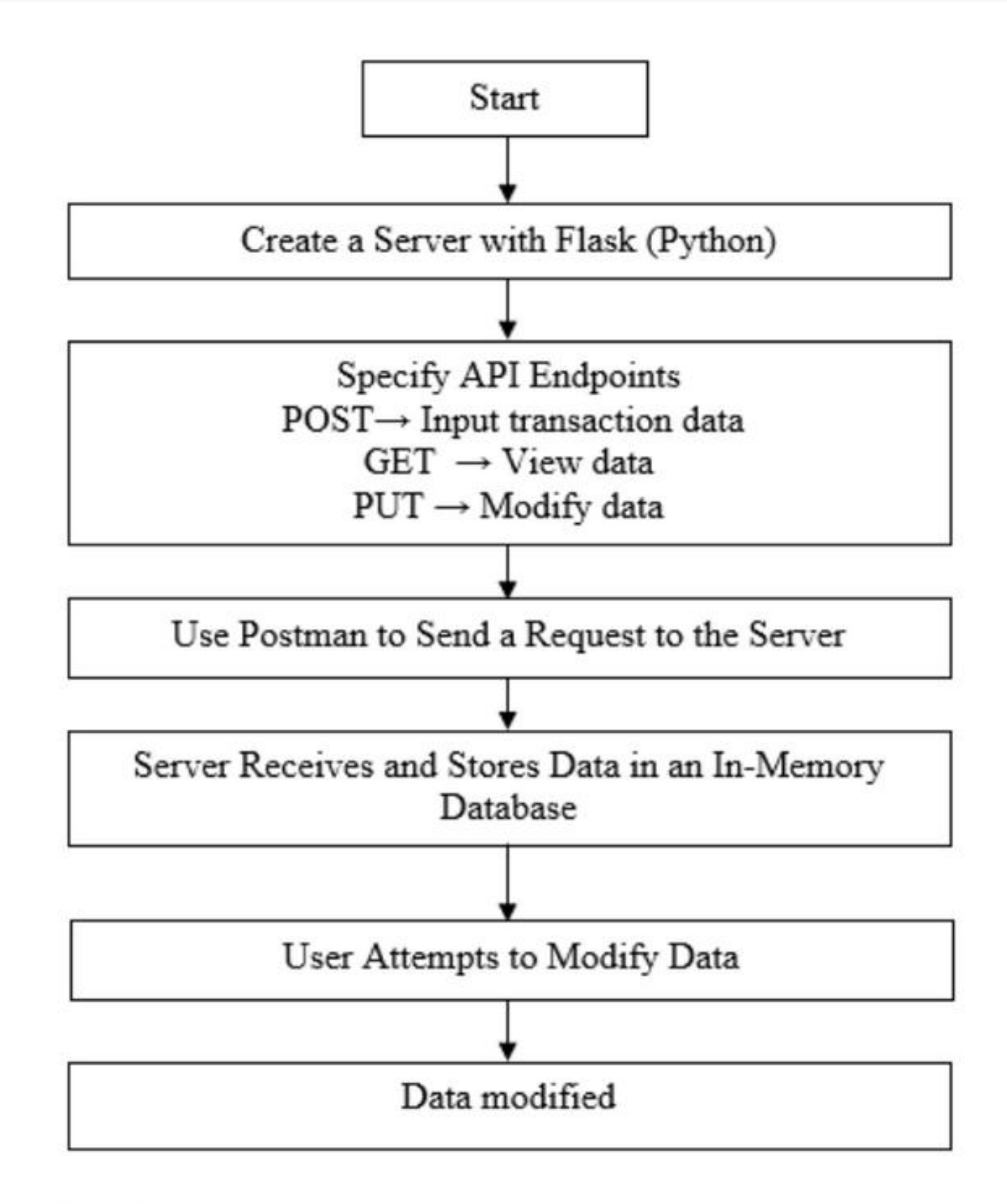

FIG 3. FLOWCHART OF SYSTEM OPTION A

1. Developed using Python Flask and RESTful APIs.
2. Data stored in a centralized in-memory database.
3. Supports data insertion, retrieval, and modification.
4. Allows data manipulation without audit trails.

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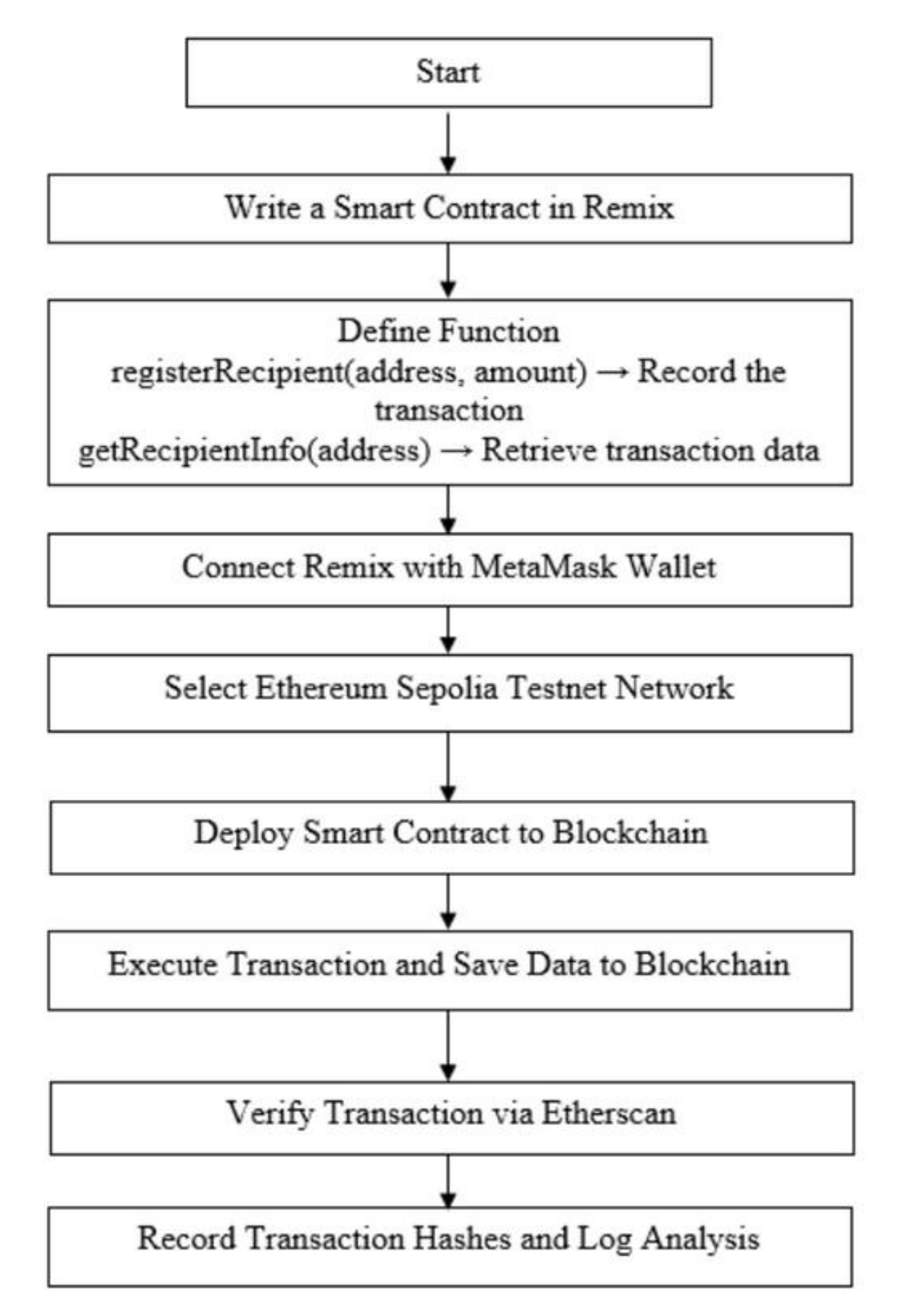
SYSTEM B: BLOCKCHAIN SYSTEM

FIG 4. FLOWCHART OF SYSTEM OPTION B



1. Implemented using Solidity smart contracts.
2. Deployed on Ethereum Sepolia Testnet.
3. Integrated with MetaMask wallet.
4. Data cannot be modified or deleted after submission.
5. Provides full transparency and immutability.

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RESULTS COMPARISON

Criterion	System A (Centralized)	System B (Blockchain)
Data Manipulation	Possible via PUT endpoint	Impossible (immutable)
Transparency	Restricted, requires server access	Publicly accessible via Etherscan
Auditability	Limited	Fully traceable
Security	Vulnerable to insider manipulation	Resistant to unauthorized change

TABLE I. OBSERVATION METRICS

1. System A:
- Data manipulation is possible.
- No public audit trail.
- High corruption risk.
1. System B:
- Data is immutable.
- Publicly verifiable transactions.
- Strong resistance to manipulation.

SYSTEM A RESULTS (EVIDENCE)

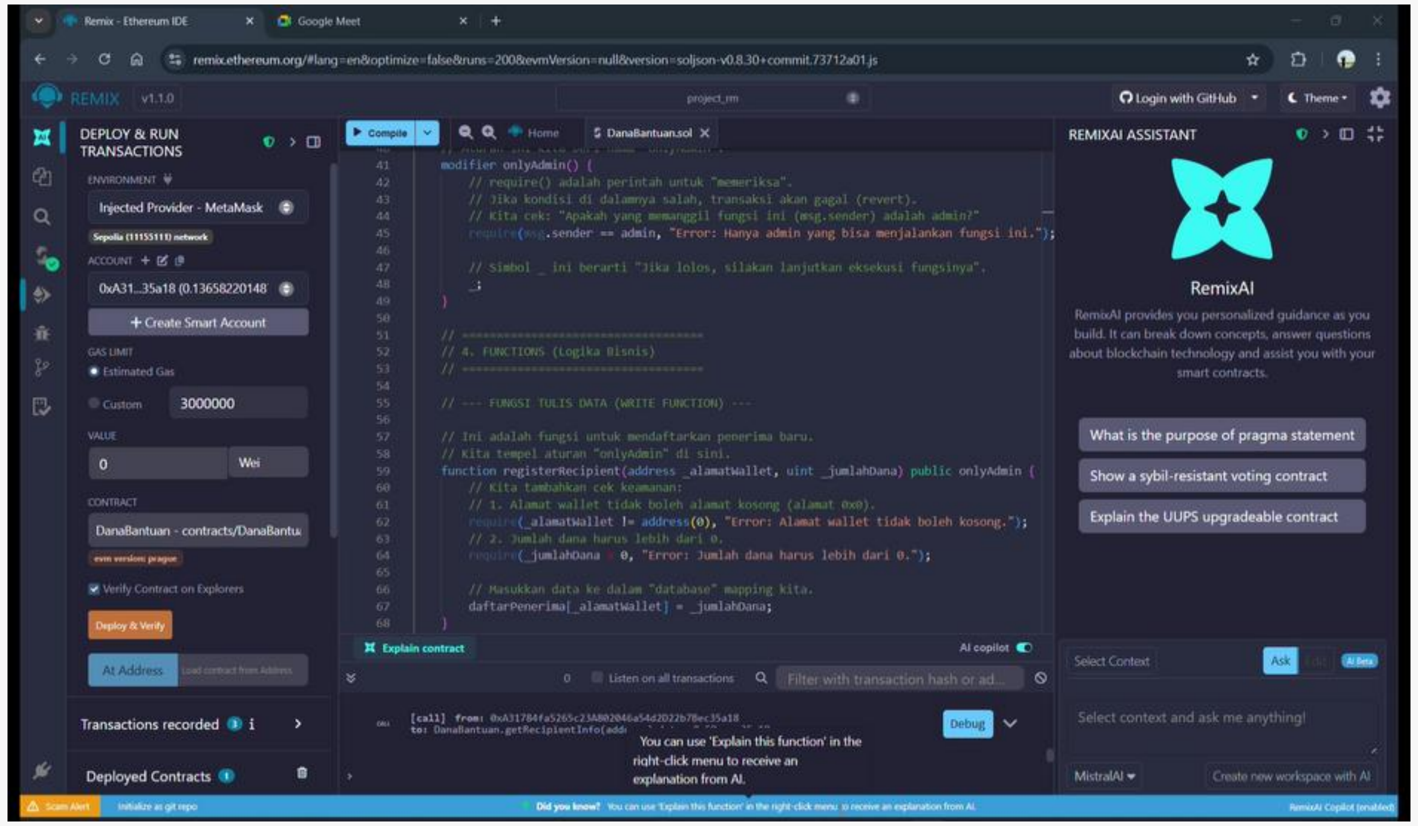

FIG 5. DISPLAY OF SYSTEM OPTION A

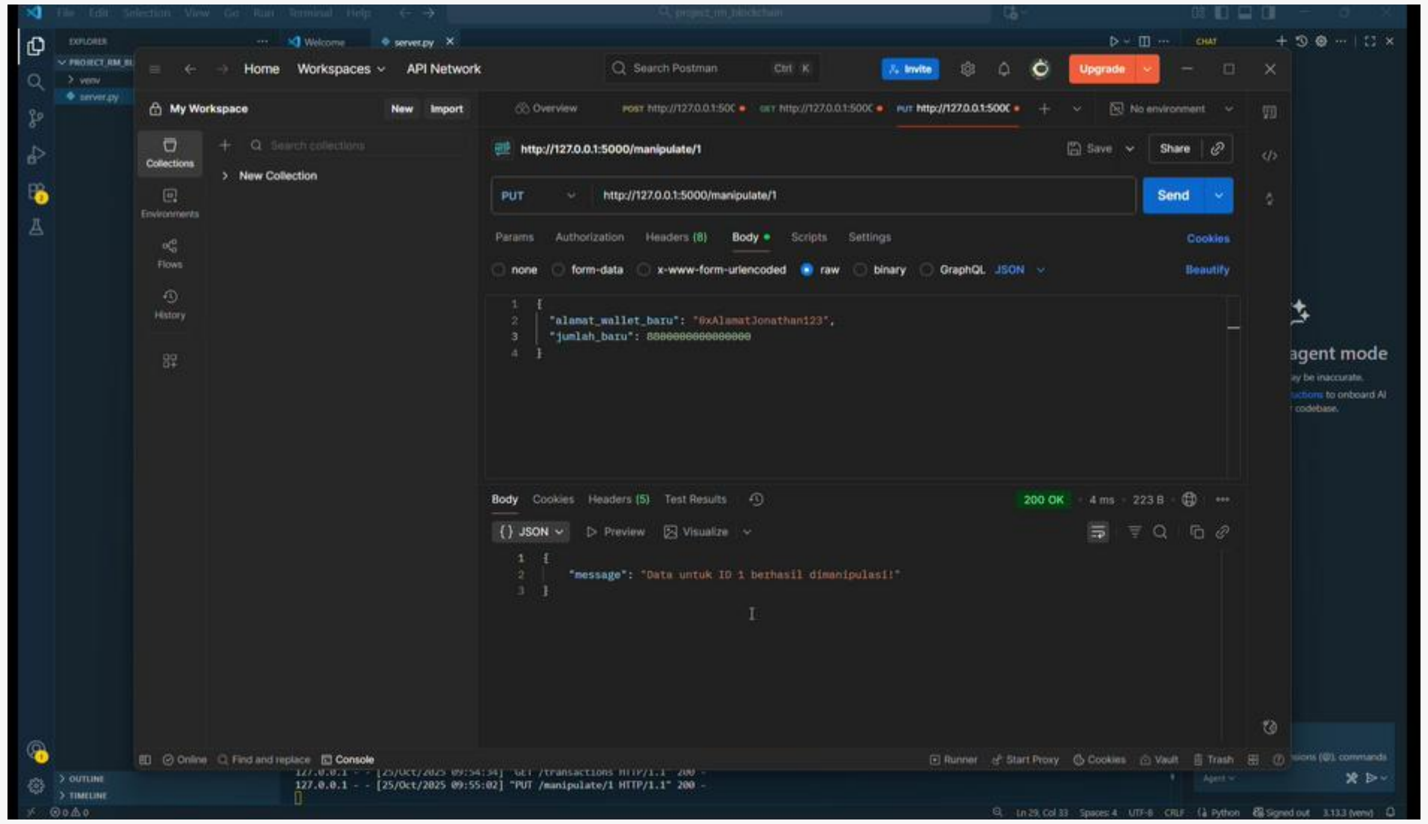

FIG 7. OPTION A DURING DATA MANIPULATION

- Data can be modified using PUT endpoint.
- Changes overwrite original records.

- No trace of data manipulation.
- Demonstrates weakness of centralized systems.

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SYSTEM B RESULTS (EVIDENCE)

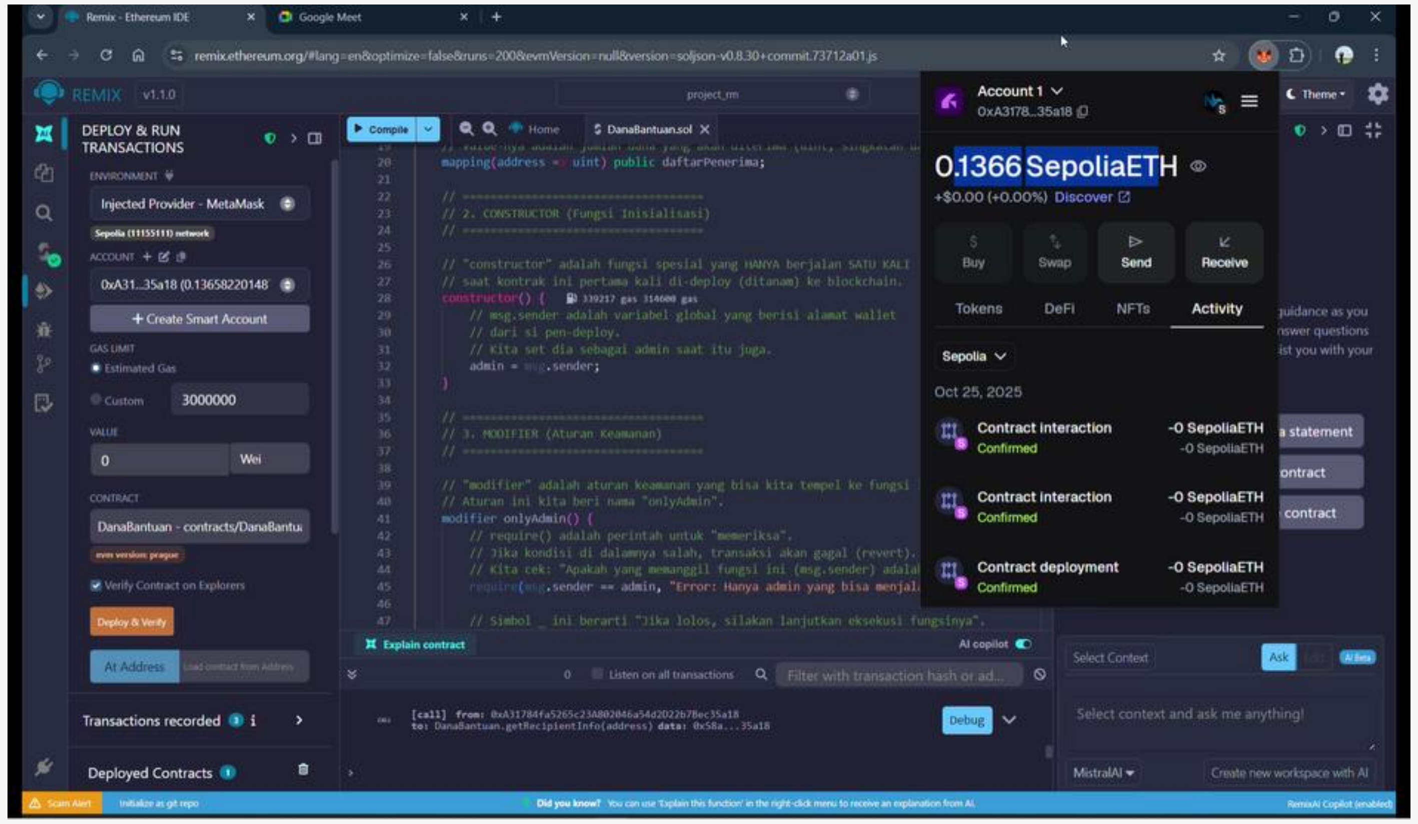

FIG 9. DISPLAY OF SYSTEM OPTION B

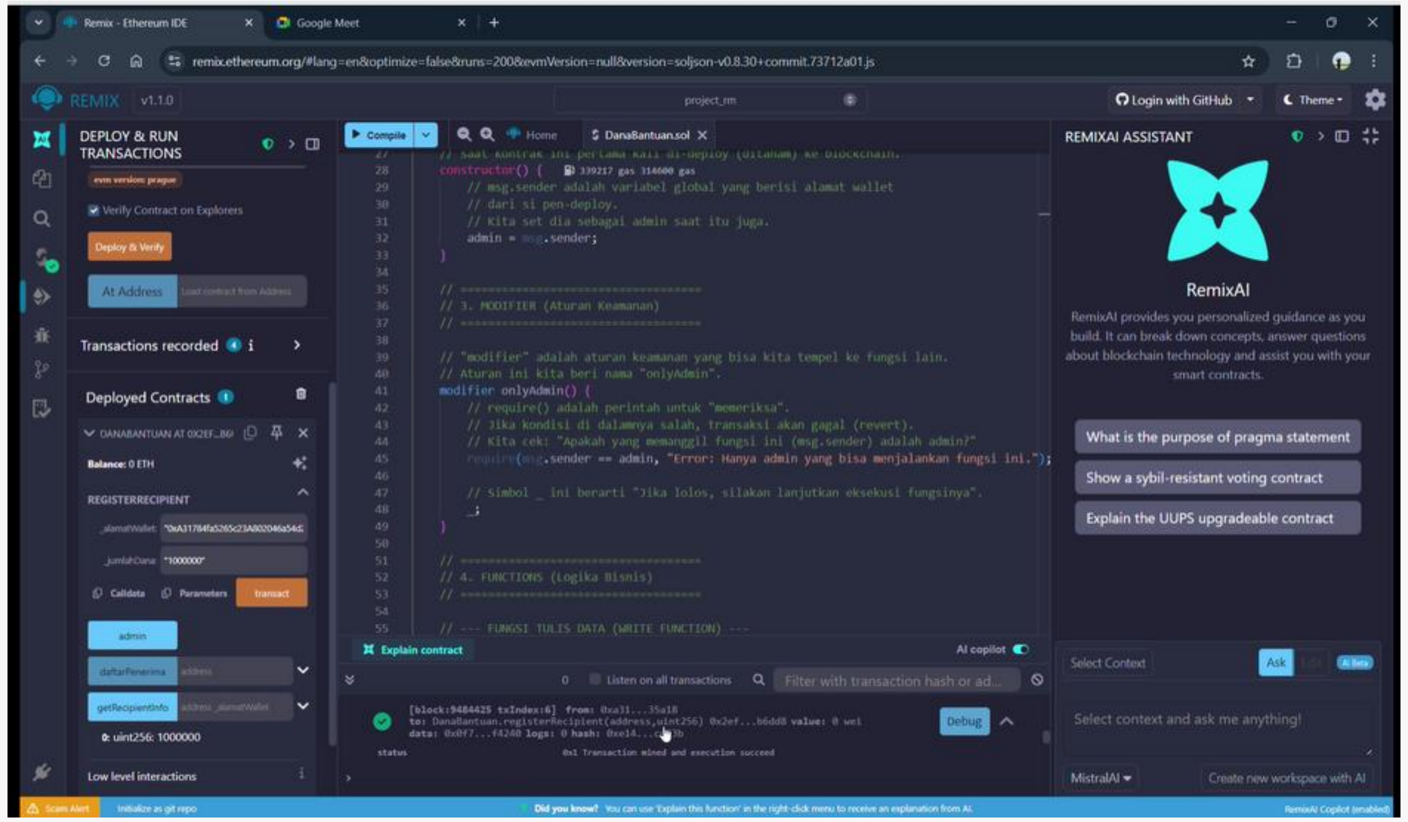

FIG 13. RECORDED TRANSACTIONS AND DATA UPDATES IN OPTION B

- Transactions require MetaMask approval.
- Every action generates a transaction hash.

- Records are publicly visible via Etherscan.
- Data integrity is fully preserved.

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CONCLUSION & RECOMMENDATION

- Blockchain-based systems outperform centralized systems.
- Immutability and transparency reduce corruption risks.
- Blockchain strengthens accountability and public trust.
- Recommended for gradual adoption in e-government systems.

THANK YOU